# Polynomial Graphs and Models



ex: $f(x) = 3\cdot(x+3)^4(x-2)^2$

A factored form makes zeros and multiplicities visible:

$$f(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots (x - r_k)^{m_k}.$$

1. Determine the degree and leading term; describe both ends.
2. Find the $x$-intercepts and the $y$-intercept $(0, f(0))$.
3. Use multiplicity to decide whether the graph crosses or touches each zero.
4. Use a test value between consecutive zeros to determine the sign there.
5. Draw a smooth curve. Check the turning-point bound and both ends.

A qualitative sketch should show the correct intercepts, signs, and end behavior.
Exact turning-point coordinates are not required in this section.

# Analyze and Sketch a Cubic



**Example.** Analyze $f(x) = (x+1)(x-2)^2$ and sketch its graph.

Find the degree and leading term, both end limits, all intercepts, each zero's multiplicity, and the maximum possible number of turning points. Show crossing or touching at each zero.

- degree is 3, leading term $2 \cdot x^3$
- odd degree, positive leading coefficient

$\lim_{x \to -\infty} f(x) = -\infty$, $\lim_{x \to \infty} f(x) = \infty$

- $f(0) = (0+1)(0-2)^2 = 1 \cdot 4 = 4$

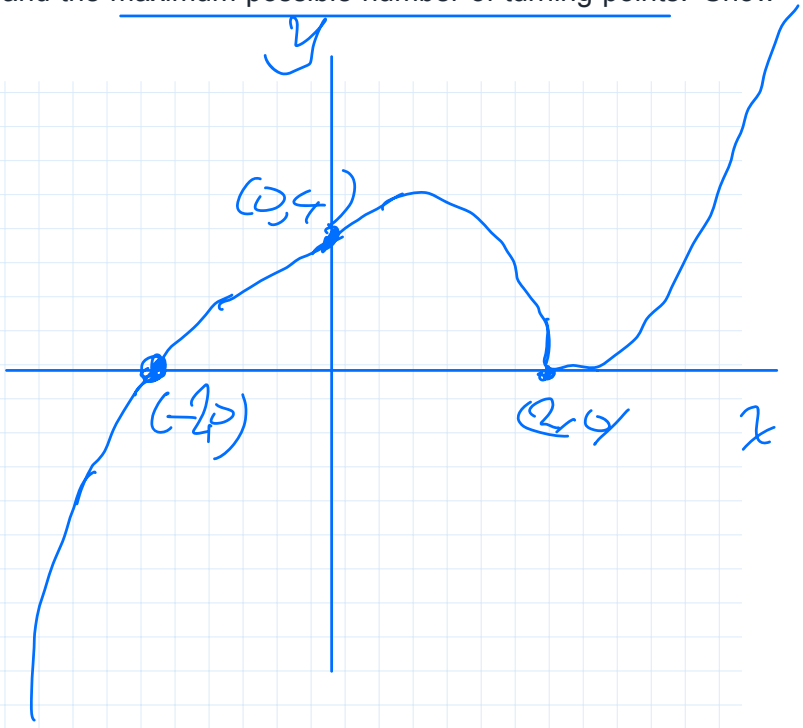
y-int: $(0, 4)$

$(x+1)(x-2)^2 = 0$

$x = -1$, $x = 2$

mult: 2

$(-1, 0)$, $(2, 0)$




at $(-1, 0)$, odd mult, so crossing

at $(2, 0)$, even mult, turning

, 2 turning pts